# Rehan Sanjay Venkatesan

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## Profile

Engineer who builds AI agents that hold up in production, and measures what they actually do. **33 fixes merged into 11 open-source codebases** I didn't write, including **12 in LiveKit Agents** and a core connection-pool fix that repaired 11 plugins at once. jambonz commissioned me to build its Pipecat integration, now part of Pipecat's core. Built **Atlas**, a ~31,000-line LLM phone agent, on my own. Looking to build agent products end to end on a small team.

## Experience

**jambonz** **Remote**
*Contract Engineer* *Oct–Nov 2026*

- Paid by jambonz to build its Pipecat serializer (pipecat #6573, merged into core), so a phone call on jambonz can drive a Pipecat voice agent. It reads each call's sample rate and audio layout from the stream itself, and rejects a layout it can't play rather than sending audio at the wrong speed.

**LiveKit Agents, Pipecat, jambonz, and vendor SDKs** **Remote**
*Open-Source Contributor* *Jul 2026–Present*

- **33 PRs merged across 11 repositories**: 12 into LiveKit Agents, reviewed by its core team, and 4 into Pipecat.
- **Core framework fix** (#7139): the shared connection pool closed sockets that were still streaming audio, and reused a connection opened with old settings. Fixed it in the pool itself, repairing all **11 plugins** that depend on it. Then found the same bug in 3 TTS plugins and fixed each (#7132, #7133, #7140).
- Removed a class of silently lost `asyncio` tasks: fixed **15 instances** and got ruff's `RUF006` rule enabled in LiveKit's CI so it cannot return (#7054).
- Fixed resources left open after shutdown, and runtime settings that were stored but never sent: LiveKit (#7023, #7012, #7050), Pipecat (#5464, #5632), and telephony call-status bugs in jambonz (#1584, #1585).
- **Vendor SDKs**: 4 merged in Gnani's Pipecat plugin (websocket reconnects, leaked sockets), 4 in Socratic's Cosmo SDK, and 2 in Varis (an unbounded key fetch, an unclamped polling interval).
- Every fix ships with a regression test, checked by reverting the fix and confirming the test fails.

**Codtech IT Solutions; Makos Infotech** **Chennai**
*Internships* *2024–2025*

Backend intern (Feb–Mar 2025): Java and MySQL modules, REST APIs, authentication. Frontend intern (Jun–Jul 2024): React components in an Agile team.

## Projects

**Atlas**
*LLM voice agent for phone calls — built solo, ~31,000 lines* *2026*

- Real-time pipeline: Twilio Media Streams → Deepgram speech-to-text → LLM (Groq) → Deepgram text-to-speech over WebSockets. NestJS, Next.js, Postgres/Prisma, Redis/BullMQ.
- A call-session state machine for dialling, speaking, listening and hang-up; barge-in stops the agent when the caller talks; a silence watchdog; an instant acknowledgement masks LLM latency.
- After each call an LLM classifies the outcome and books callbacks, with WhatsApp follow-up; outbound dialling only within legal calling hours. Tested end to end on real calls.

**Benchmarks**
*Published measurements of voice-agent behaviour* *2026*

- Turn-end latency: scored turn detection against ground truth across **265 labelled calls**. When a maintainer found errors in my scoring, I published the corrections.
- Half-open websockets (STT, TTS): tested all **39 streaming STT and TTS plugins** against a dead connection; of the 32 that could be measured, 15 never notice. The TTS fix is open as #8035.
- Plugin health: a nightly check of **75 community Pipecat plugins** against the latest release and main, which finds breakage before users do.

**InvoiceCheck.in**: Live GST invoice checker: an 11-point compliance check before marketplaces reject an invoice, OCR auto-fill, Razorpay payments. Next.js, TypeScript, Supabase.

## Skills

**Languages**: Python (asyncio, pytest), TypeScript, JavaScript, Java, SQL

**Agents & LLMs**: LiveKit Agents, Pipecat, OpenAI, Anthropic and Groq APIs, Deepgram, Twilio Media Streams

**Backend**: NestJS, Node.js, Postgres, Prisma, Redis, BullMQ, Supabase, Docker, GitHub Actions

**Frontend**: React, Next.js, Tailwind

## Education

**Sri Sairam Institute of Technology** **Chennai**
*B.E. Computer Science and Engineering, CGPA 7.24* *2022–2026*